

menu.tn

Complete SaaS Product Conception Document

Digital Menu & Business Presence Platform for Tunisia

Version 1.0 | 2025 | Confidential & Proprietary

1. Executive Summary

1.1 The Problem

Tunisia's restaurant, café, and hospitality sector is experiencing rapid digitalization pressure — yet the vast majority of small businesses remain digitally invisible. The current solutions available to them are broken at both ends:

- Traditional web agencies charge between 1,500 and 5,000 TND for a basic website, which is financially inaccessible to a small café owner.
- Imported QR menu solutions (e.g., Menupro, QR Tiger) are priced in USD/EUR, lack Arabic support, and are not adapted to Tunisian business norms.
- Social media pages (Facebook, Instagram) are used as a poor substitute but lack professional structure, are not searchable, and cannot be easily updated.
- Business owners waste hours calling or messaging to handle menu changes, pricing updates, and promotions — or pay developers each time.
- Customers scanning a QR code often land on a PDF that cannot be zoomed, is outdated, or loads poorly on mobile networks.

1.2 The Solution

menu.tn is a SaaS platform that enables any restaurant, café, salon, or small business in Tunisia to create a professional, mobile-first digital presence in under 30 minutes — without writing a single line of code and without hiring a web agency.

The platform delivers five core capabilities:

- A public-facing business page accessible at menu.tn/business-name, optimized for mobile and built in Arabic, French, and English.
- A full menu management system: categories, products, prices, photos, and descriptions — all editable in real time from any device.
- Automatic QR code generation linked to the public page, ready to print, embed, or share.
- A business dashboard to manage content, track analytics, and control subscription settings.
- A subscription model priced in Tunisian Dinars, affordable for micro and small businesses.

1.3 Market Opportunity

Tunisia has approximately 12,000 registered restaurants and cafés, with an estimated 25,000–40,000 informal food service venues across the country. Add barbershops, salons, car rentals, and other small businesses, and the total addressable market exceeds 100,000 businesses.

Even at 2% market penetration with an average of 150 TND/year, the platform generates 3,000,000 TND annually. At 5% penetration: 7.5M TND/year.

The market conditions are favorable: smartphone penetration in Tunisia exceeds 73%, mobile internet usage is growing rapidly, and digital literacy among business owners is increasing — especially among the under-40 demographic.

1.4 Competitive Advantage

Advantage	Description
Local-first design	Built specifically for Tunisia: Arabic RTL, TND pricing, local payment methods
Radical simplicity	A business owner can go live in under 30 minutes, no training required
Affordable pricing	99–149 TND/year vs. 1,500–5,000 TND for a web agency
Real-time updates	Price or product changes reflect instantly — no developer needed
QR + direct URL	Dual-access: QR scan + shareable link for WhatsApp, Instagram bio, etc.
Multi-vertical ready	Architecture supports cafés today, salons and hotels tomorrow

2. Market Analysis

2.1 International Competitors

2.1.1 Squarespace / Wix (Website Builders)

Strengths	Beautiful templates, strong SEO, global brand, multi-language
Weaknesses	Priced in USD, no Arabic RTL optimization, overkill for a simple menu, no QR ecosystem, no local payment support
Our Opportunity	Focus on simplicity and vertical specialization: menu.tn does ONE thing perfectly for ONE market

2.1.2 Menupro / Menufy / QR Tiger (QR Menu Services)

Strengths	Specialized for menus, QR-native, restaurant-focused feature set
Weaknesses	USD/EUR pricing, no Arabic support, no local payment gateways, UI not adapted for North African users, often slow on low-bandwidth networks
Our Opportunity	Full localization: Arabic + French + English, TND pricing, offline-compatible design

2.1.3 Linktree / Bento (Link-in-Bio Tools)

Strengths	Ultra-simple, shareable, popular in MENA via Instagram
Weaknesses	Not a real menu system, no product catalog, no analytics for businesses, no QR print system
Our Opportunity	Be the vertical-specific alternative: a Linktree built for menus, with real business tools behind it

2.2 North Africa & MENA Competitors

2.2.1 Zomato / Talabat (Delivery Aggregators)

Strengths	High brand recognition, large user base, delivery infrastructure
Weaknesses	Commission-based (20–30%), require delivery capability, not useful for cafés/salons, no custom page or branding for the business
Our Opportunity	We are a presence tool, not a delivery platform. No commission. Perfect for businesses that do not deliver.

2.2.2 Facebook Pages / Instagram Profiles (Social as Website)

Strengths	Free, familiar, widely used by Tunisian business owners
Weaknesses	Algorithm-dependent reach, no structured menu, hard to update prices, no QR code, no analytics, not professional, no custom URL
Our Opportunity	Educate the market: your Facebook page is marketing. menu.tn is your professional address. They complement each other.

2.3 Tunisia-Specific Landscape

As of early 2025, no established local SaaS platform serves this exact positioning in Tunisia. The digital menu and SMB presence market is effectively uncontested at the local level. Observations from the field:

- Many restaurants in Tunis, Sfax, and Sousse use WhatsApp groups to share menus as photos — an extremely low-quality experience for the customer.
- A small number of restaurants use generic PDF QR codes that link to an unbranded PDF hosted on Google Drive.
- Web agencies that serve the hospitality sector charge 2,000–5,000 TND for a basic WordPress site, then 200–500 TND/month for maintenance.
- No locally-built, Arabic-first, menu-specific SaaS platform exists with a subscription model in TND.

The competitive landscape in Tunisia represents a genuine first-mover opportunity. The key risk is not competition from local players — it is the education cost of convincing business owners to move from free (but poor) solutions to a paid (but professional) one.

3. Product Positioning

3.1 Ideal Customer Profile (ICP)

Dimension	Primary Target	Secondary Target
Business Type	Restaurant, café, fast food	Salon, barbershop, hotel
Location	Tunis, Sfax, Sousse, Monastir	Secondary cities & tourist zones
Size	1–10 employees, single location	2–20 employees, 1–3 locations
Owner Age	25–45 years old	35–55 years old
Tech Comfort	Smartphone-confident, uses WhatsApp daily	Basic smartphone user
Current Digital Presence	Facebook page only, or nothing at all	Website from agency, or Facebook page
Budget	100–400 TND/year for digital tools	400–1,000 TND/year for digital tools
Key Motivation	Look professional, attract customers	Reduce agency dependency, save money

3.2 Main Pain Points

- Pain 1: 'I cannot afford a website.' — Web agencies quote 2,000–5,000 TND. menu.tn costs 99–149 TND/year.
- Pain 2: 'My menu changes every week.' — PDF or WhatsApp updates are chaotic. menu.tn allows real-time edits from a phone.
- Pain 3: 'My customers cannot find my prices.' — No structured, accessible menu exists. menu.tn creates a permanent public page.
- Pain 4: 'I do not know how many people visit my page.' — Zero analytics with current tools. menu.tn tracks scans, views, and engagement.
- Pain 5: 'My online presence looks unprofessional.' — Facebook photos and WhatsApp screenshots feel amateurish. menu.tn provides a clean, branded page.

3.3 Unique Value Proposition

"Your restaurant is online, professional, and always up to date — in 30 minutes, for less than 150 TND per year."

3.4 Positioning Statements

For business owners:

menu.tn is the digital presence platform for Tunisian restaurants and cafés that lets you create a professional, mobile-first menu page and QR code in 30 minutes — without a web agency and without monthly developer fees.

For customers:

menu.tn gives you instant access to any restaurant or café's menu by scanning a QR code — always updated, always mobile-ready, in Arabic, French, or English.

Brand Personality:

- Simple but not simplistic
- Local but modern
- Affordable but professional
- Arabic-first but globally standard

4. MVP Definition

4.1 Must Have (Launch Blockers)

These features are non-negotiable. Without them, the product does not work and cannot be sold.

Feature	Why It's a Must Have
User registration & login (email + password)	Without authentication, no SaaS exists
Business profile creation (name, logo, cover, contact, slug)	The core identity of the business on the platform
Category management (create, edit, delete, reorder)	Menus need structure to be usable
Product management (name, description, price, photo, availability toggle)	This is the core product value proposition
Public menu page (menu.tn/slug)	Without this, there is nothing for customers to see
QR code auto-generation per business	The primary customer acquisition mechanism for users
Mobile-first responsive design	75%+ of customers will scan from a smartphone
Arabic + French language support (UI and menu content)	Non-negotiable for the Tunisian market
Free plan with limited products (5–10 items)	Needed to allow trial without payment friction
Basic subscription management (Free → Premium upgrade)	The revenue mechanism

4.2 Should Have (Next Sprint)

High value, but the MVP can ship without them. Build within 60–90 days of launch.

Feature	Rationale
Product image upload (with compression and CDN)	Increases engagement; can start with text-only menus
English language support	Important for hotels/tourist venues, less so for local cafés at launch
Basic analytics dashboard (QR scans, page views)	High perceived value; drives upgrade decisions
WhatsApp click-to-chat button on public page	Critical for the Tunisian market where WhatsApp is primary communication
Social share buttons	Organic growth mechanism
Business hours display	Customers want to know when you are open

Email notifications (welcome, subscription renewal)	Reduces churn and improves onboarding
Password reset flow	Required for production security

4.3 Nice To Have (Phase 2+)

Valuable features that differentiate the premium offering but are not needed to validate the market.

Feature	Rationale
Custom domain support (myrestaurant.com)	Required for Business plan; complex to implement
Multiple staff users per account	Needed for hotels and multi-location businesses
Product variants (size, spice level, extras)	High complexity; most cafés do not need this at launch
Advanced analytics (heatmaps, peak hours, top products)	Valuable but not Day 1 critical
Online ordering / reservations	Major scope increase; separate product consideration
Printable menu PDF export	Nice for dual-use (print + digital)
Google Maps integration	Helps customers find the business
Review / rating system	Social proof feature; requires moderation
API access for integrations	Enterprise feature; too early for MVP
Multi-location management	Needed for chains; not Day 1 target audience

5. User Roles & Permissions

5.1 Super Admin (Platform Operator)

The internal team managing the platform. Access to all data, all tenants, and all system settings.

Permission	Actions	Restrictions
Full platform access	View, edit, delete any account, business, or content	Should use separate admin domain (admin.menu.tn)
User management	Create/suspend/delete user accounts, reset passwords, impersonate users for support	All actions logged with timestamp and admin ID
Subscription management	Override plan, apply discounts, extend trial periods, manually activate/suspend accounts	Financial overrides require dual-admin confirmation
Analytics access	View platform-wide analytics: total users, MRR, churn rate, top businesses by traffic	PII data masked in export reports
Content moderation	Review and remove inappropriate business content, flag suspicious accounts	Must document reason for removal
System configuration	Manage pricing plans, feature flags, global settings, email templates	Infrastructure changes require deployment pipeline

5.2 Business Owner (Account Admin)

The primary user. Created their account, owns the business profile, manages all content, and controls the subscription.

Permission	Actions	Restrictions
Account management	Update email, password, notification preferences, delete account	Cannot delete account with active paid subscription
Business profile	Edit name, logo, cover, description, contact info, slug, hours	Slug change sends redirect but may affect QR codes — show warning
Menu management	Create/edit/delete categories and products, reorder items, toggle product availability	Free plan limited to 10 products; prompt upgrade if limit reached
Media uploads	Upload logo, cover image, product photos	Max 5MB per image; automatically compressed
QR code	View, download (PNG/SVG), regenerate QR code	Cannot set QR to point to external URL

Analytics	View own page analytics: scans, views, top products, traffic by day	Cannot see data of other businesses
Subscription	View current plan, upgrade/downgrade, view invoice history, cancel	Downgrade takes effect at end of billing period
Staff management	Add/remove staff users, set permissions (Business plan only)	Cannot grant staff Super Admin permissions

5.3 Staff Member

An employee granted limited access by the Business Owner. Can update menu content but cannot touch billing or settings.

Permission	Actions	Restrictions
Menu content	Edit product prices, descriptions, availability; add new products	Cannot delete categories created by Owner
Media	Upload product images	Cannot change logo or cover image
Read-only access	View analytics dashboard	Cannot change any subscription or billing settings
Restrictions	Cannot access billing, subscription, or account deletion	Cannot add or remove other staff members

5.4 Customer (Public User)

The end-consumer scanning the QR code or visiting the public page. No account required. No authentication.

Action	Details
View public business page	Access menu.tn/business-name — no login required
Browse menu categories and products	View names, descriptions, prices, and images
Click WhatsApp / phone / social links	Direct contact with business via external apps
Share the page	Share link via copy, WhatsApp, or social media buttons
Switch language (if enabled)	Toggle between Arabic, French, English on the public page
No write access	Cannot place orders, submit reviews, or modify any content (MVP scope)

6. User Journey Maps

6.1 Business Owner — Onboarding Journey

Step	Action	System Response	Potential Drop-off Point
1. Discovery	Sees ad on Facebook/Instagram, hears about platform from a friend, or searches for 'menu QR Tunisie'	Landing page loads in <2 seconds on 3G	High — if landing page is slow or unclear
2. Registration	Clicks 'Start for Free', enters name + email + phone + password	Account created, welcome email sent, redirected to onboarding wizard	Medium — form must be <5 fields
3. Business Profile	Enters business name, chooses category (restaurant/café/etc.), uploads logo	Slug auto-generated from business name (editable); live preview shown	Low if preview is engaging
4. Menu Creation	Creates first category (ex: Boissons), adds first product with name and price	Product appears in live preview on right side of screen	Low — this is the 'aha moment'
5. QR Generation	System automatically generates QR code linked to menu.tn/their-slug	QR preview shown; download button (PNG/SVG/PDF) prominently displayed	Very low — this is exciting
6. Publish & Share	Clicks 'My page is live!' confirmation; shares link on WhatsApp	Public page accessible; sharing modal opens with WhatsApp, copy link options	Low
7. Subscription Prompt	After setup, modal appears explaining Free plan limits (10 products)	Upgrade CTA shown: '99 TND/year — unlimited menu, analytics, and more'	High — this is the conversion moment
8. Ongoing Use	Returns to dashboard weekly to update prices, add seasonal items, check analytics	Dashboard shows recent scan count and top product as a motivating hook	Monitor churn at 30/60/90 days

6.2 Customer — Scan Journey

Step	Action	Experience Goal
------	--------	-----------------

1. QR Scan	Customer points phone camera at QR code on table, counter, or printed menu	Instant redirect — page must start loading in <1 second
2. Business Page Load	Public page loads with cover photo, logo, business name, hours	Visual impact: feel of a professional restaurant website
3. Browse Categories	Scrolls through category tabs (Entrées, Plats, Boissons, Desserts)	Clear navigation; sticky category bar stays visible while scrolling
4. View Products	Taps a product to see full description, photo, price	Clean product card with image, name, description, price in TND
5. Contact Action	Taps 'WhatsApp us' or 'Call' button	Opens WhatsApp/phone app directly — no friction
6. Share	Taps share button to send link to a friend	Native share sheet opens

7. Database Design

7.1 Entity Relationship Overview

Multi-tenant shared database with tenant isolation enforced via `business_id` foreign key on all tenant-scoped tables. PostgreSQL is the primary database.

7.2 Core Tables

users

Column	Type / Description
id	UUID PRIMARY KEY — universally unique, non-guessable
email	VARCHAR(255) UNIQUE NOT NULL — login identifier
password_hash	VARCHAR(255) NOT NULL — bcrypt hash, never store plaintext
full_name	VARCHAR(255) NOT NULL
phone	VARCHAR(20) — optional, used for WhatsApp contact
role	ENUM('super_admin', 'business_owner', 'staff') DEFAULT 'business_owner'
email_verified_at	TIMESTAMP — null until verification email clicked
last_login_at	TIMESTAMP — used for churn detection
created_at	TIMESTAMP DEFAULT NOW()
updated_at	TIMESTAMP DEFAULT NOW() ON UPDATE

businesses

Column	Type / Description
id	UUID PRIMARY KEY
owner_id	UUID REFERENCES users(id) ON DELETE CASCADE
slug	VARCHAR(100) UNIQUE NOT NULL — used in menu.tn/slug URL
name	VARCHAR(255) NOT NULL — business display name
name_ar	VARCHAR(255) — Arabic name (optional)
category	ENUM('restaurant', 'cafe', 'fastfood', 'salon', 'barbershop', 'hotel', 'other')
description	TEXT — about the business
description_ar	TEXT — Arabic description

logo_url	VARCHAR(500) — CDN URL for logo image
cover_url	VARCHAR(500) — CDN URL for cover banner
phone	VARCHAR(20)
whatsapp	VARCHAR(20) — may differ from phone
address	VARCHAR(500)
city	VARCHAR(100)
is_published	BOOLEAN DEFAULT FALSE — false until owner completes setup
primary_color	VARCHAR(7) DEFAULT '#1B4F72' — hex color for branding
created_at / updated_at	TIMESTAMP

business_hours

Column	Type / Description
id	UUID PRIMARY KEY
business_id	UUID REFERENCES businesses(id) ON DELETE CASCADE
day_of_week	SMALLINT 0–6 (0=Sunday, 1=Monday, ..., 6=Saturday)
open_time	TIME — e.g., '08:00'
close_time	TIME — e.g., '23:00'
is_closed	BOOLEAN DEFAULT FALSE — for days the business is closed

categories

Column	Type / Description
id	UUID PRIMARY KEY
business_id	UUID REFERENCES businesses(id) ON DELETE CASCADE
name	VARCHAR(255) NOT NULL — e.g., 'Boissons Chaudes'
name_ar	VARCHAR(255) — Arabic category name
sort_order	INTEGER DEFAULT 0 — drag-to-reorder position
is_active	BOOLEAN DEFAULT TRUE
created_at / updated_at	TIMESTAMP

products

Column	Type / Description
id	UUID PRIMARY KEY

business_id	UUID REFERENCES businesses(id) — denormalized for fast queries
category_id	UUID REFERENCES categories(id) ON DELETE SET NULL
name	VARCHAR(255) NOT NULL
name_ar	VARCHAR(255) — Arabic name
description	TEXT
description_ar	TEXT
price	DECIMAL(10,3) NOT NULL — TND supports 3 decimal places
image_url	VARCHAR(500) — CDN URL
is_available	BOOLEAN DEFAULT TRUE — toggle on/off without deleting
is_featured	BOOLEAN DEFAULT FALSE — featured badge on public page
sort_order	INTEGER DEFAULT 0
created_at / updated_at	TIMESTAMP

subscriptions

Column	Type / Description
id	UUID PRIMARY KEY
business_id	UUID REFERENCES businesses(id) ON DELETE CASCADE
plan	ENUM('free', 'premium', 'business') DEFAULT 'free'
status	ENUM('active', 'grace_period', 'suspended', 'cancelled') DEFAULT 'active'
started_at	TIMESTAMP
expires_at	TIMESTAMP — null for free plan
grace_period_ends_at	TIMESTAMP — 7-day grace after expiry
payment_reference	VARCHAR(255) — reference from payment gateway
payment_method	VARCHAR(100) — e.g., 'card', 'virement', 'cash'
amount_paid	DECIMAL(10,3) — amount in TND
created_at / updated_at	TIMESTAMP

analytics_events

Column	Type / Description
id	BIGSERIAL PRIMARY KEY — integer for performance at scale
business_id	UUID REFERENCES businesses(id)
product_id	UUID REFERENCES products(id) ON DELETE SET NULL — nullable

event_type	ENUM('page_view', 'qr_scan', 'product_view', 'contact_click', 'share_click')
session_id	UUID — anonymous session tracking (no PII)
country	VARCHAR(2) — ISO code, derived from IP geolocation
device_type	ENUM('mobile', 'tablet', 'desktop')
referrer	VARCHAR(500) — where the visitor came from
created_at	TIMESTAMP DEFAULT NOW()

qr_codes

Column	Type / Description
id	UUID PRIMARY KEY
business_id	UUID REFERENCES businesses(id) UNIQUE — one QR per business
redirect_url	VARCHAR(500) — menu.tn/slug
svg_url	VARCHAR(500) — CDN URL for downloadable SVG
png_url	VARCHAR(500) — CDN URL for downloadable PNG
scan_count	INTEGER DEFAULT 0 — denormalized counter for quick reads
created_at / updated_at	TIMESTAMP

8. System Architecture

8.1 Architecture Overview

The recommended architecture is a modern monolith with a clear separation between frontend, backend API, and storage layers. Start as a monolith; extract microservices only when specific scaling bottlenecks emerge.

8.2 Technology Stack

Layer	Technology	Justification
Frontend (Dashboard)	Next.js 14 (React) + Tailwind CSS	SSR for SEO, fast page loads, great DX, large ecosystem, Arabic RTL support via CSS logical properties
Frontend (Public Pages)	Next.js 14 with SSG/ISR	Static generation for blazing-fast public pages; ISR (Incremental Static Regeneration) for real-time menu updates
Backend API	FastAPI (Python 3.11+)	Native async support handles concurrent QR scans efficiently; auto-generated Swagger docs at /docs; Pydantic validation built-in; production-grade and used by Uber, Netflix, and Microsoft
API Validation	Pydantic v2	Automatic request/response validation and serialization; schemas serve as the contract between frontend and backend; zero extra code for input sanitization
ORM & Migrations	SQLAlchemy 2.0 (async) + Alembic	Async-native ORM prevents event loop blocking; Alembic handles schema migrations cleanly across environments
Background Tasks	Celery + Redis	Offloads CPU-bound tasks (image compression, QR generation, email sending) to async workers; prevents blocking the API event loop
Auth & Security	python-jose (JWT) + passlib + bcrypt	Industry-standard JWT handling; bcrypt with salt rounds=12 for password hashing; matches security spec exactly
Admin Panel	sqladmin (FastAPI-compatible)	Django-admin-like interface for Super Admin operations (manage businesses, override subscriptions) — no need to build a separate admin UI
Database	PostgreSQL 15 on managed service	ACID compliance, JSON support for flexible menu schemas, excellent performance, full-text search for future product search feature
Cache	Redis (via Upstash or self-hosted)	Cache public menu pages (invalidate on content update), rate limiting, Celery broker, session storage

File Storage	Cloudflare R2 or AWS S3	Cheap object storage for images; R2 has zero egress fees (important for cost control in Tunisia); accessed via boto3 (S3-compatible)
Image Processing	Pillow (Python)	Server-side image compression and resizing before upload; run inside Celery worker to avoid blocking the API
CDN	Cloudflare (free tier)	Image delivery, DDoS protection, Tunisia latency reduction via Cloudflare's Casablanca/Cairo edge nodes
Email	Resend.com or AWS SES	Transactional email (welcome, invoices, renewal reminders); low cost, high deliverability; Python SDKs available for both
Hosting	Railway.app or Render.com (startup phase)	Simple deployment for FastAPI (Dockerfile or Nixpacks), managed PostgreSQL, autoscale; migrate to AWS/Hetzner at scale
QR Generation	qrcode (Python library)	Native Python library; generates SVG and PNG on demand inside Celery worker; no external dependency or API call needed
Analytics Ingestion	Custom events table + Postgres aggregations	Avoid third-party analytics cost for MVP; query analytics_events with daily/weekly aggregation queries via SQLAlchemy async

8.3 Scalability Plan

Scale Stage	Users	Architecture Adjustment
Seed / MVP	0–500 businesses	Single Railway/Render instance, shared DB, all-in-one Next.js app
Early Growth	500–5,000 businesses	Separate API server from frontend, add Redis cache for public pages, CDN image delivery
Growth	5,000–50,000 businesses	Database read replicas, Celery + Redis queue for async tasks (QR gen, email, image processing), horizontal API scaling with Gunicorn + Uvicorn workers
Scale	50,000+ businesses	Consider microservices for analytics and QR generation, multi-region deployment, dedicated analytics DB (Clickhouse)

8.4 Estimated Monthly Hosting Costs

Service	Startup (0–500 users)	Growth (5,000 users)
App Hosting (Railway/Render)	~\$20/month	~\$80/month
PostgreSQL (managed)	~\$15/month	~\$60/month
Redis Cache (Upstash)	~\$0 (free tier)	~\$20/month

File Storage (R2/S3)	~\$5/month	~\$30/month
CDN (Cloudflare)	\$0 (free tier)	\$0–\$20/month
Email (Resend)	~\$0 (free tier 3k/mo)	~\$20/month
Domain + SSL	~\$15/year	~\$15/year
TOTAL	~\$55/month	~\$210/month

9. API Design

9.1 API Conventions

- Base URL: `https://api.menu.tn/v1`
- All responses: JSON with envelope: `{ success: bool, data: {}, error: {code, message} }`
- Authentication: Bearer JWT token in Authorization header
- Rate limiting: 100 requests/min per IP for public endpoints; 500/min for authenticated
- Pagination: cursor-based using `?cursor=&limit=` params

9.2 Authentication Endpoints

Meth od	Endpoint	Auth	Description
POST	<code>/auth/register</code>	None	Register new user. Body: {email, password, full_name, phone}. Returns: {user, token}
POST	<code>/auth/login</code>	None	Login. Body: {email, password}. Returns: {user, token, expires_at}
POST	<code>/auth/logout</code>	Required	Invalidate current token
POST	<code>/auth/forgot-password</code>	None	Send reset email. Body: {email}
POST	<code>/auth/reset-password</code>	None	Body: {token, new_password}
GET	<code>/auth/me</code>	Required	Return current authenticated user profile
PATCH	<code>/auth/me</code>	Required	Update profile: name, phone, notification preferences

9.3 Business Management

Meth od	Endpoint	Auth	Description
POST	<code>/businesses</code>	Required	Create new business profile. One per owner (Free/Premium). Body: {name, slug, category}
GET	<code>/businesses/me</code>	Required	Get own business profile with full details
PATCH	<code>/businesses/me</code>	Required	Update business details: name, description, contact, hours, colors
DELETE	<code>/businesses/me</code>	Required	Delete business and all associated data (requires password confirmation)
GET	<code>/businesses/:slug/public</code>	None	Public endpoint — returns full menu data for public page render

POST	/businesses/me/logo	Required	Upload logo image (multipart/form-data). Returns CDN URL
POST	/businesses/me/cover	Required	Upload cover image. Returns CDN URL

9.4 Menu Management

Method	Endpoint	Auth	Description
GET	/categories	Required	List all categories for the authenticated business
POST	/categories	Required	Create category. Body: {name, name_ar, sort_order}
PATCH	/categories/:id	Required	Update category name or sort order
DELETE	/categories/:id	Required	Delete category (products become uncategorized)
POST	/categories/reorder	Required	Update sort_order for multiple categories. Body: [{id, sort_order}]
GET	/products	Required	List all products (with optional ?category_id= filter)
POST	/products	Required	Create product. Checks plan limit before insert.
PATCH	/products/:id	Required	Update product details
DELETE	/products/:id	Required	Soft delete (is_active = false) or hard delete
POST	/products/:id/image	Required	Upload product image
PATCH	/products/:id/availability	Required	Toggle is_available. Body: {is_available: bool}

9.5 QR Code & Analytics

Method	Endpoint	Auth	Description
GET	/qr	Required	Get QR code for own business (auto-creates if not exists)
POST	/qr/regenerate	Required	Regenerate QR code (same URL, new visual)

GET	/analytics/overview	Required	Returns: total scans, total views, top products, daily trend (last 30 days)
GET	/analytics/events	Required	Paginated list of events for own business (Premium+)
POST	/analytics/track	None	Public endpoint — track customer events. Rate-limited. Body: {business_id, event_type, product_id?, session_id}

9.6 Subscription Management

Method	Endpoint	Auth	Description
GET	/subscription	Required	Get current subscription status, plan, expiry, and feature limits
POST	/subscription/upgrade	Required	Initiate upgrade. Returns payment intent or redirect URL
POST	/subscription/cancel	Required	Cancel subscription (takes effect at period end)
GET	/subscription/invoices	Required	List past payments
POST	/subscription/webhook	None (signature)	Payment gateway webhook — update subscription status on payment confirmed

10. Dashboard UX Design

10.1 Navigation Structure

The dashboard is a single-page application with a left sidebar navigation on desktop and a bottom tab bar on mobile. All screens are accessible within 2 taps/clicks from the home screen.

- Home / Overview
- Menu — Categories + Products sub-pages
- Business Settings — Profile, Hours, Appearance
- QR Code
- Analytics (Premium)
- Subscription
- Account Settings

10.2 Onboarding Wizard

Shown to new users who have not yet completed setup. A 4-step progress bar guides them through:

1. Step 1 — Business Info: Name, category, city, phone, WhatsApp number
2. Step 2 — Look & Feel: Upload logo, upload cover image, choose primary color
3. Step 3 — Your First Menu: Create one category, add one product with price
4. Step 4 — Go Live: See the QR code and public page URL; download QR; share on WhatsApp

Skip button available at each step. Business is auto-published once onboarding wizard is completed.

10.3 Dashboard Home

Component	Description
Welcome Banner	Personalized greeting; shows business name and completion status if setup incomplete
Quick Stats Row	3 cards: QR Scans Today Page Views This Week Total Products
Live Preview Link	Button: 'View your page' opens menu.tn/slug in new tab
Recent Activity	Last 5 analytics events (for Premium users)
Upgrade Banner	Shown to Free users: animated CTA showing benefits of Premium
Quick Actions	4 shortcut buttons: Add Product Download QR Edit Profile View Analytics

10.4 Menu Management Page

Component	Description
Category Tabs	Horizontal scrollable tabs; each tab shows product count badge
Category Actions	+ Add Category button (top right); each tab has Edit and Delete icons
Product List	Card grid (2 columns on desktop, 1 on mobile); each card shows image, name, price, availability toggle
Product Card Actions	Edit (pencil icon), Delete (trash icon), Drag handle for reordering
Add Product Button	Floating action button (FAB) in bottom right; opens slide-in panel
Product Form Panel	Slide-in from right: Name (FR), Name (AR), Description, Price (TND), Image upload, Availability toggle
Plan Limit Banner	Shown to Free users when at limit: 'You have 10/10 products. Upgrade to add more.'

10.5 QR Code Page

Component	Description
QR Preview	Large, centered QR code display with business name below it
URL Display	Shows full URL: menu.tn/business-slug (copyable with one click)
Download Options	3 buttons: Download PNG (for digital sharing) Download SVG (for print) Download PDF (A4, print-ready)
Print Tips	Collapsible section: recommended print sizes, how to add to table stands, etc.
Scan Counter	Shows total all-time QR scans
Regenerate Button	Accessible via 'More Options'; warns user that old printed QR codes will still work (same URL)

10.6 Analytics Page (Premium)

Component	Description
Date Range Picker	Presets: Today, Last 7 Days, Last 30 Days, Last 3 Months; custom range option
Summary Cards	Total QR Scans Total Page Views Unique Sessions Average Daily Views
Line Chart	Views and scans over time — dual-line chart; identifies peaks (weekend dinner rush, etc.)
Top Products	Bar chart: 5 most-viewed products; useful for menu optimization

Device Breakdown	Donut chart: Mobile vs. Tablet vs. Desktop (always dominated by mobile)
Traffic Sources	Table: Direct QR Scan WhatsApp Share Instagram Other
Export Button	Download CSV of raw events (Business plan only)

11. Public Page Design (menu.tn/business-name)

11.1 Page Architecture

The public page is the product that customers actually see. It must be beautiful, fast, and completely functional without JavaScript enabled (progressive enhancement). Target: <1.5s FCP on a 4G connection from Tunisia.

11.2 Page Layout Sections

Section	Content	Notes
Hero / Header	Cover image (full-width, 40vh), logo (overlapping bottom edge), business name, city, category badge	Image lazy-loaded; placeholder color from dominant color extraction
Contact Bar	Phone icon WhatsApp icon Location pin — all tappable; opens respective app	Fixed at top on mobile (sticky); critical for conversion
Status Badge	Open Now (green) / Closed (red) based on business hours and current time	Computed server-side; displayed prominently
Category Navigation	Horizontal scrollable tab bar with category names; sticky below contact bar	Clicking a category smoothly scrolls to that section
Menu Sections	Each category as a titled section; products displayed in responsive card grid	2 columns on mobile; 3 columns on tablet+
Product Cards	Image (if uploaded), product name, description (truncated, expandable), price in TND	Tapping card opens a modal with full details and full-size image
Footer	Platform branding: 'Powered by menu.tn'; social media icons; copyright	Small, unobtrusive; includes 'Create your own menu' CTA

11.3 Mobile Experience Requirements

- Minimum tap target: 44×44px for all interactive elements
- Text size minimum: 16px to prevent iOS auto-zoom on input fields
- Touch-optimized: swipe gestures for category navigation
- Image optimization: WebP format served via Cloudflare, sized to device viewport
- Offline indicator: graceful degradation message if user has no connection
- No horizontal scroll: layout must fit all screen widths from 320px

11.4 Performance Targets

Metric	Target	Method
First Contentful Paint (FCP)	< 1.5 seconds on 4G	Next.js SSG/ISR, Cloudflare CDN edge delivery
Largest Contentful Paint (LCP)	< 2.5 seconds	Preload hero image; optimize cover image to <200KB WebP
Cumulative Layout Shift (CLS)	< 0.1	Reserve image dimensions in HTML; avoid late-loading layout shifts
Time to Interactive (TTI)	< 3 seconds on 3G	Minimal JavaScript bundle; server-rendered HTML as primary content
Page weight	< 500KB for first load	Image compression, code splitting, remove unused CSS

12. QR Code System

12.1 QR Generation Flow

- 5. Business owner completes setup and publishes their page.
- 6. Backend auto-generates QR code pointing to: `https://menu.tn/{slug}`
- 7. qrcode library generates SVG and PNG versions (512×512px, 1024×1024px).
- 8. Files uploaded to Cloudflare R2 storage at `/qr/{business_id}/{version}/`.
- 9. URLs stored in `qr_codes` table.
- 10. Dashboard displays QR immediately with download options.

12.2 URL & Slug Management

Topic	Policy / Implementation
Slug format	Lowercase letters, numbers, hyphens only. Max 50 characters. Auto-generated from business name on creation.
Slug uniqueness	Checked against businesses table on creation; suffix -2, -3 added if taken (e.g., le-petit-cafe-2).
Slug changes	Owner can change slug from settings. Old slug stored in <code>slug_history</code> table and redirects via 301 to new URL.
Reserved slugs	Blocklist maintained: 'admin', 'api', 'www', 'login', 'register', 'blog', 'about', 'contact', etc.
Custom domain (Business plan)	DNS CNAME record points to menu.tn. Platform handles SSL via Let's Encrypt. Reverse proxy maps custom domain to business slug.

12.3 QR Code Tracking

The QR code URL includes an analytics tracking parameter that is stripped before the customer sees it:

QR URL format: `https://menu.tn/slug?ref=qr` → Server logs QR scan event → Redirect 302 to `https://menu.tn/slug` (clean URL)

This allows distinguishing QR scans from direct URL visits in analytics, without exposing tracking parameters to customers.

12.4 Print & Download Formats

Format	Use Case	Specifications
PNG (512px)	Digital sharing via WhatsApp, social media	Transparent background option; white background option

PNG (1024px)	High-resolution digital use	Full color, menu.tn branding below QR
SVG	Scalable vector for professional printing	Infinite resolution; ideal for print shops
PDF (A4)	Ready-to-print table card	QR centered on A4 page with business name, logo, and 'Scan to see our menu' text in Arabic and French

13. Subscription & Billing System

13.1 Plan Comparison

Feature	Free	Premium (99–149 TND/yr)	Business (249–399 TND/yr)
Products in menu	10 items	Unlimited	Unlimited
Categories	3	Unlimited	Unlimited
Product images	No	Yes	Yes
Custom logo & cover	Yes	Yes	Yes
Analytics dashboard	No	Basic (30 days)	Advanced (12 months)
QR code download	PNG only	PNG + SVG + PDF	PNG + SVG + PDF
WhatsApp button	Yes	Yes	Yes
Custom domain	No	No	Yes
Staff users	1 (owner only)	1 (owner only)	Up to 5
Multi-location	No	No	Up to 3
CSV export	No	No	Yes
Priority support	No	Email	Phone + Email
'Powered by' badge	Visible	Removable	Removable
Branding colors	No	Yes	Yes

13.2 Billing Logic

Scenario	System Behavior
New subscription	Payment collected; subscription.started_at = now(); expires_at = +365 days; status = 'active'
Approaching expiry	Email reminder sent 30 days, 7 days, and 1 day before expires_at
Expiry	Status set to 'grace_period'; grace_period_ends_at = +7 days; page remains live
Grace period end	If not renewed: status = 'suspended'; public page shows 'Currently unavailable' message; owner dashboard shows suspension notice
Renewal	Same as new subscription; starts from original expires_at (not renewal date) to not penalize early renewers
Upgrade mid-cycle	Immediate feature unlock; no proration in MVP (add proration in Phase 2)
Downgrade	Takes effect at period end; features remain until then

Cancellation	Subscription continues until expires_at; then moves to Free plan (does not delete data)
Account reactivation	Owner pays; status = 'active'; page goes live immediately

13.3 Tunisian Payment Methods

As of 2025, Tunisia has limited online payment infrastructure. The platform must support both digital and manual payment flows.

Method	Status	Implementation Notes
Flouci (Tunisian fintech)	Recommended — Phase 1	Popular mobile payment app in Tunisia; has merchant API; accepts bank cards
Bank card (Visa/Mastercard)	Phase 1	Via Flouci or Stripe (Stripe available in Tunisia since 2023)
Virement bancaire (Bank transfer)	Phase 1 manual	Owner sends transfer; admin manually confirms in dashboard; subscription activated
Cash payment (in-person)	Phase 1 manual	For early adopters; admin records payment in dashboard
D17 / PayTic	Phase 2	Local fintech alternatives gaining adoption
SOBFLOUS	Phase 2	Mobile payment used in retail

14. Security Architecture

14.1 Authentication & Authorization

Concern	Implementation
Password storage	bcrypt with salt rounds = 12. Never log or transmit plaintext passwords.
JWT tokens	Short-lived access tokens (15 minutes) + long-lived refresh tokens (30 days) stored in httpOnly cookies. Never store JWT in localStorage.
Session management	Refresh token rotation: each use issues a new refresh token and invalidates the old one.
Multi-tenant isolation	Every API endpoint that queries tenant data validates that the authenticated user's business_id matches the requested resource. Enforced at service layer, not just controller.
Email verification	Required before accessing dashboard. Verified via signed token in email link.
Password reset	Token valid for 1 hour only. Single-use. Stored as hash in DB, not plaintext.
Admin access	Separate admin subdomain (admin.menu.tn). Admin tokens have additional claim; verified separately from owner tokens.

14.2 API Security

Concern	Implementation
Rate limiting	100 req/min per IP for public endpoints. 500/min for authenticated. 10 req/min for auth endpoints (login, register). Return 429 with Retry-After header.
Input validation	All inputs validated server-side using Pydantic v2 schemas. Whitelist approach enforced automatically by FastAPI route handlers. Whitelist, not blacklist.
SQL injection	Parameterized queries only. Never string-concatenate user input into SQL. Use ORM (Prisma or Drizzle).
XSS prevention	All user-generated content escaped on render. Content Security Policy headers set on public pages.
File upload security	Validate MIME type server-side (not just extension). Strip EXIF metadata from images. Max file size enforced before processing.
CORS	Restrict CORS to known frontend domains only. No wildcard origin in production.
HTTPS	Enforce HTTPS everywhere. HSTS header with max-age=31536000. No HTTP fallback.

14.3 Data Protection & Backups

- Daily automated PostgreSQL backups retained for 30 days.
- Point-in-time recovery (PITR) enabled on managed DB provider.
- Images backed up to secondary storage bucket weekly.
- GDPR-adjacent principles applied (even though Tunisia's data protection law is less strict): users can request data export and deletion.
- Business data isolated per tenant; no cross-tenant data leakage by design.

15. Multi-Tenant Architecture

15.1 Strategy: Shared Database, Separate Schemas Approach

For a SaaS platform at this scale, the recommended approach is a shared database with tenant isolation enforced at the application layer — also known as the 'shared schema' pattern.

Pattern	Description	Recommended For
Shared DB, Shared Tables	All tenants in same tables; business_id FK isolates data. Enforced at app layer.	✓ Recommended for menu.tn MVP to Phase 3
Shared DB, Separate Schemas	Each tenant gets own Postgres schema. Better isolation, more complex migrations.	Consider at 10,000+ tenants if compliance requires it
Separate DB per Tenant	Maximum isolation. Operational nightmare at scale.	Enterprise/high-compliance only

15.2 Isolation Enforcement

- Every DB query for tenant data includes WHERE business_id = :current_user_business_id.
- Middleware layer extracts business_id from JWT and injects into request context before any controller runs.
- ORM-level scoping: create a base repository class that automatically scopes all queries to current tenant.
- Row-level security (RLS) in PostgreSQL as an additional safety net — even if application code is buggy, DB will reject cross-tenant reads.

15.3 Scalability Considerations

- Business ID indexed on all tenant-scoped tables for O(log n) lookups.
- Analytics events table partitioned by month after 1M rows.
- Connection pooling via PgBouncer to handle thousands of concurrent connections.
- Read replicas for analytics queries to avoid impacting write performance.

16. Analytics System

16.1 Event Tracking Design

All analytics are first-party. No Google Analytics or Meta Pixel on the public page (to maximize performance and avoid ad-blocker issues). Events are tracked via a lightweight endpoint.

Event Type	Trigger	Data Captured
qr_scan	User arrives via ?ref=qr parameter	business_id, timestamp, device_type, country
page_view	Any page visit (including direct/WhatsApp link)	business_id, timestamp, device_type, country, referrer
product_view	Customer taps to expand a product card	business_id, product_id, timestamp
contact_click	Customer taps WhatsApp, phone, or map button	business_id, contact_type, timestamp
share_click	Customer uses share button	business_id, platform, timestamp
category_view	Customer clicks a category tab	business_id, category_id, timestamp

16.2 Dashboard Metrics for Business Owners

Metric	Plan	Calculation
QR Scans (Today / Week / Month)	Free + Premium	COUNT(events) WHERE event_type='qr_scan' AND business_id=X
Page Views (Today / Week / Month)	Free + Premium	COUNT(events) WHERE event_type='page_view' AND business_id=X
Unique Sessions	Premium	COUNT(DISTINCT session_id) WHERE business_id=X
Top 5 Products by Views	Premium	GROUP BY product_id ORDER BY count DESC LIMIT 5
Traffic by Day (line chart)	Premium	GROUP BY DATE(created_at) for selected date range
Device Breakdown	Premium	GROUP BY device_type
Contact Click Rate	Premium	contact_clicks / page_views * 100
Peak Hour Analysis	Business	GROUP BY HOUR(created_at) — identify busiest hours
Monthly Trend Comparison	Business	Compare current month vs previous month

17. Product Roadmap

Phase 1 — MVP (Months 0–3)

Goal: Get 50 paying businesses on the platform. Validate the core loop: sign up → create menu → share QR. Keep it simple.

Feature	Priority	Effort
User registration, login, password reset	P0	Medium
Business profile creation with slug	P0	Medium
Logo and cover image upload	P0	Medium
Category and product CRUD	P0	Medium
Product availability toggle	P0	Low
Auto-generated QR code (PNG download)	P0	Low
Public menu page (mobile-first, FR + AR)	P0	High
Free plan limits (10 products)	P0	Low
Basic subscription (Premium plan)	P0	Medium
Flouci / bank transfer payment (manual confirmation MVP)	P0	Medium
Dashboard: product management, QR download	P0	High
Onboarding wizard (4 steps)	P1	Medium
WhatsApp + phone buttons on public page	P1	Low
Email notifications (welcome, renewal reminder)	P1	Low

Phase 2 — Growth (Months 3–6)

Goal: Reach 200 paying businesses. Improve product quality, add analytics, launch referral program.

Feature	Priority	Effort
Analytics dashboard (scans, views, top products)	P0	High
QR code SVG + PDF download	P0	Low
Business hours display on public page	P0	Low
English language support	P1	Medium

Referral program (refer a business = 1 month free)	P1	Medium
Product image uploads with CDN delivery	P0	High
Stripe card payment integration	P0	Medium
Social share buttons on public page	P1	Low
Google Maps / Waze link on public page	P1	Low
Business plan launch (staff users, advanced analytics)	P1	High
Public landing page (menu.tn homepage) with SEO	P1	High
Slug change with redirect	P1	Low
Brand color customization	P1	Low

Phase 3 — Scale (Months 6–12)

Goal: 500+ businesses, expand to hotels and salons, explore partnerships.

- Custom domain support for Business plan
- Multi-location business management
- Vertical expansion: service menus for salons, barbershops, car rentals
- Product variants (size, options, extras)
- Printable PDF menu export
- Automated renewal billing and receipts
- Franchise/agency dashboard (manage multiple clients)
- API access for integrations (POS systems, delivery apps)
- Mobile app (React Native) for dashboard on-the-go

Phase 4 — Market Expansion (Year 2)

- Algeria and Morocco market entry (French-first, different payment providers)
- Online ordering integration (table booking, takeaway)
- Loyalty program features for business owners
- AI-powered features: auto-translate menu to Arabic/English, product description suggestions
- White-label version for web agencies to resell

18. Financial Model

18.1 Revenue Model Assumptions

- Average Revenue Per User (ARPU): 130 TND/year (blended Premium/Business)
- Free-to-paid conversion rate: 15% (conservative for Year 1)
- Monthly churn rate: 3% (industry benchmark for SMB SaaS)
- Pricing: Premium at 129 TND/year; Business at 299 TND/year
- 70% of paying customers on Premium; 30% on Business plan

18.2 Revenue Projections

Scale	Total Businesses	Paying Businesses	Avg ARPU (TND)	Annual Revenue (TND)	Monthly Revenue (TND)
Seed	100	15	130	1,950	163
Early	500	90	135	12,150	1,013
Growth	1,000	200	145	29,000	2,417
Scale	5,000	1,200	155	186,000	15,500
Mature	10,000	2,800	160	448,000	37,333

At 5,000 total businesses with 24% conversion: 186,000 TND/year (~62,000 USD at current rates). This is a viable business with a small team of 2–3 people.

18.3 Cost Structure

Cost Category	Monthly (100 users)	Monthly (1,000 users)	Monthly (5,000 users)
Hosting (Railway/Render)	160 TND (~\$55)	480 TND (~\$160)	1,200 TND (~\$400)
Database (managed Postgres)	45 TND	150 TND	450 TND
Storage + CDN (Cloudflare R2)	15 TND	90 TND	300 TND
Email service (Resend)	0 TND (free)	60 TND	120 TND
Payment processing (~2.5%)	4 TND	30 TND	385 TND
Founder salary/opportunity cost	0–2,000 TND	2,000–5,000 TND	5,000–10,000 TND
Marketing / ads budget	200–500 TND	500–1,500 TND	2,000–5,000 TND
TOTAL OPEX (excl. salary)	~420 TND	~1,310 TND	~6,455 TND

18.4 Break-Even Analysis

At 129 TND/year Premium plan: break-even (excluding founder salary) at approximately 40 paying businesses.

At 299 TND/year Business plan: break-even at approximately 17 paying businesses.

This is a low-capital business. The primary investment is development time. With 50 paying customers in the first 6 months, the platform covers its hosting costs and is technically profitable at the infrastructure level.

18.5 Development Cost Estimate

Approach	Timeline	Cost (TND)	Notes
Solo founder (full-stack)	3–4 months	0 (sweat equity)	Feasible with Next.js + Node.js; highest risk is time-to-market
2 developers (co-founders)	2–3 months	0 (sweat equity)	Recommended; one frontend-focused, one backend-focused
Freelance team (Tunisia)	2–3 months	8,000–18,000 TND	Variable quality; need senior oversight
Outsourced agency (Tunisia)	3–4 months	15,000–30,000 TND	Higher quality guarantee but costly for early stage

19. Risk Analysis

19.1 Business Risks

Risk	Probability	Impact	Mitigation
Low willingness to pay: owners expect free tools (Facebook)	High	High	Lead with free plan; focus ROI messaging ('get more customers'); in-person sales for early adopters
Slow adoption: digital literacy barrier in secondary cities	Medium	Medium	Start in Tunis, Lac, La Marsa; use referral program; partner with food industry events
High churn after Year 1 renewal	Medium	High	Track engagement; send value reminders showing QR scan counts; proactive renewal outreach 45 days before
Copy by a larger local player or agency	Low (Year 1), Medium (Year 2)	High	Move fast, build brand loyalty, accumulate reviews; moat is in product quality and local brand trust
Payment collection failures (local infra)	High	Medium	Accept manual bank transfers as backup; work with Flouci early; offer installment option

19.2 Technical Risks

Risk	Probability	Impact	Mitigation
Performance issues on 3G/4G networks in Tunisia	Medium	High	Aggressive image compression; CDN edge delivery; bundle size budget enforced in CI
Arabic RTL rendering bugs	High (early)	Medium	Use CSS logical properties from Day 1; test on real devices; hire Arabic-speaking QA
Data loss due to infrastructure failure	Low	Critical	Daily backups; managed DB provider; test restore procedure quarterly
Security breach / data leak	Low	Critical	Follow OWASP Top 10; external security audit before launch; bug bounty program in Phase 2
Slug collision or QR redirect failure	Low	High	Comprehensive test suite for slug generation; monitor 404 rates on QR redirect endpoint
Vendor lock-in (Railway, R2)	Low	Medium	Use standard protocols (S3-compatible for storage); containerize with Docker from start

19.3 Market Risks

Risk	Probability	Impact	Mitigation
------	-------------	--------	------------

Tunisian economic instability reducing SMB spending	Medium	Medium	Price below psychological threshold; offer annual plan at effective price of 10 TND/month
Dinar devaluation increasing USD-denominated hosting costs	Medium	Low-Medium	Most costs are USD-denominated; hedge by adjusting TND pricing annually; target 5× coverage ratio
Large international player enters Tunisia (Square, etc.)	Low (2 years)	High	Build local brand and community moat before they arrive; focus on features they won't localize
Regulatory changes around digital payments in Tunisia	Low	High	Maintain manual payment fallback; follow BCT (Banque Centrale de Tunisie) regulations; consult local lawyer

20. Final Recommendation & Startup Advisory

20.1 Overall Viability Assessment

Verdict: **VIABLE** — with important caveats

menu.tn addresses a real, underserved problem in a market with no direct local competitor. The technology is proven, the development cost is low, and the business model is sound. However, the primary risk is NOT technical — it is commercial. The hardest part will be convincing Tunisian business owners to pay for something they currently do for free (badly) on Facebook.

20.2 Difficulty Assessment

Dimension	Difficulty (1–5)	Notes
Technical development (MVP)	3/5	Standard SaaS architecture; nothing exotic. Main challenge is Arabic RTL and performance optimization.
Product design / UX	3/5	Must be simpler than competitors, not just equal. Onboarding is critical.
Payment integration (Tunisia)	4/5	Local payment infrastructure is immature; plan for manual fallback.
Sales & customer acquisition	4/5	Hardest part. Requires in-person sales initially. Free plan essential to reduce friction.
Retention & reducing churn	4/5	Year 1 renewal is the make-or-break moment. Must show ongoing value via analytics.
Scaling from 100 to 5,000 users	2/5	Architecture handles this well. Operational scaling is manageable.

20.3 Best Launch Strategy

Step 1: Build in 10 Weeks (Not 6 Months)

Resist feature creep. Build **ONLY** the Must Have list from Section 4. The perfect is the enemy of the launched. A working product with 10 features beats a perfect product that ships in 8 months.

Step 2: Recruit 10 Pilot Businesses Before Launch

Before you write a line of code, talk to 20 restaurant and café owners. Offer 5–10 of them 1 year free in exchange for feedback and a testimonial. This gives you real users on Day 1, a list of testimonials for social proof, and a stress test of your onboarding.

Step 3: Launch With a Face, Not Just a Website

In Tunisia, trust is personal. The founder should be the face of menu.tn. Appear in Reels and TikTok videos demonstrating the product in real restaurants. Walk into cafés in Lac, La Marsa, or downtown Tunis and offer a live demo. This direct sales approach will outperform paid ads in Year 1.

Step 4: Price Confidently

129 TND/year is 10.75 TND/month. A coffee. Do not apologize for your pricing. Position it as 'less than one hour of a web developer' or 'the cost of two cups of coffee per month for a professional digital presence.' Never discount below 99 TND/year.

Step 5: Make the Free Plan Your Best Salesperson

Every Free plan business is a walking billboard. Every customer who scans a QR code and sees 'Powered by menu.tn' is a potential lead. Make the free plan good enough to be genuinely useful — the limit should be on quantity (10 products), not quality (it must look professional).

20.4 Critical Success Factors

- Arabic-first is non-negotiable. If your dashboard and public page do not look great in Arabic, you will lose 60%+ of your potential market.
- Speed on mobile is a feature. Tunisian mobile networks are inconsistent. A page that loads in 1 second on 3G will win over one that loads in 5 seconds on 4G.
- The Year 1 renewal rate will define the business. If fewer than 50% renew, the unit economics break. Build a renewal campaign: 45 days before expiry, send personalized email showing their QR scan count for the year.
- Build a community, not just a product. A WhatsApp group or Facebook group for menu.tn subscribers creates switching costs, peer referrals, and a feedback loop.
- Your first 50 customers are your co-founders. Treat them that way. Weekly check-ins, immediate bug fixes, personal support. They are building the product with you.

20.5 Final Honest Assessment

This is a solid, fundable idea with a clear path to profitability. The technical challenge is medium. The business challenge is hard. The market exists and is underserved. The founder's biggest asset is not the code — it is their network in the Tunisian hospitality sector and their ability to build trust with small business owners. Go build it.

— End of Document —